Course Title :- Design and analysis of algorithms Laboratory

Course Code :- DSC5-Lab

Part A

1. Sort a given set of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.
2. Implement a Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted. The elements can be read from a file or can be generated using the random number generator.
3. Write a program to sort a list of N elements using Selection Sort Technique.
4. Write a program to perform Travelling Salesman Problem.
5. Write program to implement Dynamic Programming algorithm for the 0/1 Knapsack problem.
6. Write a program to find minimum and maximum value in an array using divide and conquer.
7. Write a test program to implement Divide and Conquer Strategy. Eg: Quick sort algorithm for sorting list of integers in ascending order.
8. Write a program to implement Merge sort algorithm for sorting a list of integers in ascending order.

Part B

1. Write a program to Sort a given set of n integer elements using Merge Sort method and compute its time complexity. Run the program for varied values of n> 5000, and record the time taken to sort.
2. Write a program to Sort a given set of n integer elements using Quick Sort method and compute its time complexity. Run the program for varied values of n> 5000 and record the time taken to sort
3. Write C program that accepts the vertices and edges for a graph and stores it as an adjacency matrix.
4. Write a program to Implement a function to print In-Degree, Out-Degree and to display that adjacency matrix.
5. Write a program to implement the backtracking algorithm for the sum of subsets problem.
6. Write program to implement greedy algorithm for job sequencing with deadlines
7. Write program to implement Dynamic Programming algorithm for the Optimal Binary Search Tree Problem.
8. Write a program that implements Prim's algorithm to generate minimum cost spanning Tree